

# **Bonneville Power Administration**

## **Permit-Required Confined Space Program**

**Version 1**

**9/14/2020**

**Program Manager: Landon McCowan**

**Approved by:** JOHN CANNON Digitally signed by JOHN CANNON  
Date: 2020.09.14 14:01:05 -07'00' **Date:** 9/14/2020

**Unique ID: 420-601**



## Table of Contents

|     |                                                             |    |
|-----|-------------------------------------------------------------|----|
| 1.  | Introduction .....                                          | 3  |
| 1.1 | Background.....                                             | 3  |
| 1.2 | Purpose.....                                                | 3  |
| 1.3 | Applicability/Scope.....                                    | 3  |
| 1.4 | Term, Definitions and Acronyms.....                         | 4  |
| 1.5 | Authorizing Policies .....                                  | 6  |
| 1.6 | Roles and Responsibilities .....                            | 6  |
| 2.  | Identification of Permit-Required Confined Spaces .....     | 9  |
| 2.1 | Policy .....                                                | 9  |
| 2.2 | Identifying Permit-Required Confined Spaces .....           | 9  |
| 2.3 | Controlling Access to Permit-Required Confined Spaces ..... | 10 |
| 3.  | Permit-Required Confined Space Entry Planning .....         | 10 |
| 3.1 | Process Map .....                                           | 10 |
| 3.2 | Designation of a Competent Person .....                     | 10 |
| 3.3 | Conducting the PRCS Hazard Analysis.....                    | 10 |
| 3.4 | Determining Hazard Controls.....                            | 11 |
| 3.5 | Determining Space Designation .....                         | 11 |
| 4.  | Non-Permit Confined Space Entry .....                       | 13 |
| 4.1 | Non-Permit Entry Planning .....                             | 13 |
| 4.2 | Non-Permit Entry .....                                      | 13 |
| 4.3 | Non-Permit Entry Termination.....                           | 13 |
| 5.  | Alternate Entry Process .....                               | 13 |
| 5.1 | Alternate Entry Planning .....                              | 13 |
| 5.2 | Alternate Entry .....                                       | 14 |
| 5.3 | Alternate Entry Termination .....                           | 15 |
| 6.  | Permit-Required Confined Space Entry .....                  | 15 |
| 6.1 | Permit-Required Confined Space Entry Planning .....         | 15 |
| 6.2 | Permit Required Confined Space Entry.....                   | 17 |

|                           |                                   |                                                                |  |                             |                  |
|---------------------------|-----------------------------------|----------------------------------------------------------------|--|-----------------------------|------------------|
| Organization<br><b>NF</b> |                                   | Title/Subject<br><b>Permit Required Confined Space Program</b> |  | Unique ID<br><b>420-601</b> |                  |
| Owner<br><b>CSO</b>       | Approved by<br><b>John Cannon</b> | Date<br><b>9/14/2020</b>                                       |  | Version<br><b>V1</b>        | Page<br><b>2</b> |

|     |                                                                    |    |
|-----|--------------------------------------------------------------------|----|
| 6.3 | PRCS Termination .....                                             | 18 |
| 7.  | References .....                                                   | 18 |
| 7.1 | Regulations .....                                                  | 18 |
| 7.2 | Standards.....                                                     | 18 |
| 7.3 | Training.....                                                      | 19 |
| 8.  | Review.....                                                        | 19 |
| 8.1 | Review Timeline .....                                              | 19 |
| 9.  | Document Change Control .....                                      | 19 |
| 10. | Confined Space Assessment/Permit Required Confined Space Form..... | 20 |

## 1. Introduction

### 1.1 Background

Confined spaces - such as manholes, crawl spaces, and tanks - are not designed for continuous occupancy and are difficult to exit in the event of an emergency. People working in confined spaces may face life-threatening hazards including toxic substances, electrocutions, explosions, and asphyxiation. Broad terms, describe the intent, rationale or context for the program. Bonneville Power Administration considers all confined spaces to be considered Permit-Required Confined Spaces (PRCS) and subject to the policies and procedures of this document.

### 1.2 Purpose

The purpose of this document is to prescribe the methodology in which Bonneville Power Administration will conduct work when accessing Permit-Required Confined Spaces. This document aligns the agency with Occupational Safety and Health Administration (OSHA) regulations and aims to prevent and/or eliminate injuries and illnesses occurring as part of PRCS work.

### 1.3 Applicability/Scope

This program applies to all BPA Employees and supplemental labor personnel who are conducting work that involves accessing a Permit Required Confined Space. Contractor

|                           |                                   |                                                                |                             |                  |
|---------------------------|-----------------------------------|----------------------------------------------------------------|-----------------------------|------------------|
| Organization<br><b>NF</b> |                                   | Title/Subject<br><b>Permit Required Confined Space Program</b> | Unique ID<br><b>420-601</b> |                  |
| Owner<br><b>CSO</b>       | Approved by<br><b>John Cannon</b> | Date<br><b>9/14/2020</b>                                       | Version<br><b>V1</b>        | Page<br><b>3</b> |

PRCS requirements are contained within applicable contract documents and are managed and implemented by the appropriate Contracting Officer.

#### 1.4 Term, Definitions and Acronyms

| Term                                   | Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Attendant                              | An individual stationed outside one or more permit spaces who monitors the authorized entrants and who performs all attendant's duties assigned in the employer's permit space program.                                                                                                                                                                                                                                                                                                                                           |
| Authorized Entrant                     | An employee who is authorized to enter a permit space.                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Competent Person (CP)(Confined Spaces) | A person with thorough knowledge of OSHA's Confined Space Standard, 29 CFR 1910.146 and 1926 Subpart AA, designated in writing by the employer to be responsible for the immediate supervision, implementation and monitoring of the confined space program, who through training, knowledge and experience in confined space entry is capable of identifying, evaluating and addressing existing and potential confined space hazards and, who has the authority to take prompt corrective measures with regard to such hazards. |
| Confined Space (CS)                    | A space that meets the following criteria: <ol style="list-style-type: none"> <li>1. Is large enough and so configured that an employee can bodily enter and perform assigned work; and</li> <li>2. Has limited or restricted means for entry or exit; and</li> <li>3. Is not designed for continuous employee occupancy.</li> </ol>                                                                                                                                                                                              |
| Elimination (hazard)                   | Hazard elimination is a hazard control strategy based on completely removing a material or process causing a hazard. Elimination is the most effective of the five members of the hierarchy of hazard controls in protecting workers, and where possible should be implemented before all other control methods.                                                                                                                                                                                                                  |
| Entry Supervisor                       | Person responsible for determining if acceptable entry conditions are present at a PRCS where entry is planned, for authorizing entry and overseeing entry operations, and for terminating entry.                                                                                                                                                                                                                                                                                                                                 |

|                           |                                   |                                                                |  |                             |                  |
|---------------------------|-----------------------------------|----------------------------------------------------------------|--|-----------------------------|------------------|
| Organization<br><b>NF</b> |                                   | Title/Subject<br><b>Permit Required Confined Space Program</b> |  | Unique ID<br><b>420-601</b> |                  |
| Owner<br><b>CSO</b>       | Approved by<br><b>John Cannon</b> | Date<br><b>9/14/2020</b>                                       |  | Version<br><b>V1</b>        | Page<br><b>4</b> |

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hazardous Atmosphere                  | <p>An atmosphere that may expose employees to the risk of death, incapacitation, and impairment of ability to self-rescue (that is, escape unaided from a permit space), injury, or acute illness from one or more of the following causes:</p> <ul style="list-style-type: none"> <li>• Flammable gas, vapor, or mist in excess of 10% of its lower explosive limit (LEL);</li> <li>• Airborne combustible dust at a concentration that meets or exceeds its LEL. This concentration may be approximated as a condition in which the dust obscures vision at a distance of 5 feet or less.</li> <li>• Atmospheric oxygen concentration below 19.5% or above 23.5%;</li> <li>• Atmospheric concentration of any substance for which a dose or a permissible exposure limit is published in Subpart G, Occupational Health and Environmental Control, or in Subpart Z, Toxic and Hazardous Substances, of this Part and which could result in employee exposure in excess of its dose or permissible exposure limit. An atmospheric concentration of any substance that is not capable of causing death, incapacitation, and impairment of ability to self-rescue, injury, or acute illness due to its health effects is not covered by this provision.</li> <li>• Any other atmospheric condition that is immediately dangerous to life or health. For air contaminants for which OSHA has not determined a dose or permissible exposure limit, other sources of information, such as Safety Data Sheets that comply with the Hazard Communication Standard, section 1910.1200 of this Part, published information, and internal documents can provide guidance in establishing acceptable atmospheric conditions.</li> </ul> |
| Permit-Required Confined Space (PRCS) | <p>A confined space that has one or more of the following characteristics:</p> <ul style="list-style-type: none"> <li>• Contains or has a potential to contain a hazardous atmosphere;</li> <li>• Contains a material that has the potential for engulfing an entrant;</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

|                    |                            |                                                         |                   |                      |           |
|--------------------|----------------------------|---------------------------------------------------------|-------------------|----------------------|-----------|
| Organization<br>NF |                            | Title/Subject<br>Permit Required Confined Space Program |                   | Unique ID<br>420-601 |           |
| Owner<br>CSO       | Approved by<br>John Cannon |                                                         | Date<br>9/14/2020 | Version<br>V1        | Page<br>5 |

|                              | <ul style="list-style-type: none"><li>Has an internal configuration such that an entrant could be trapped or asphyxiated by inwardly converging walls or by a floor which slopes downward and tapers to a smaller cross-section; or</li><li>Contains any other recognized serious safety or health hazard.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                              |                                               |                                              |                                               |                 |    |                   |                   |        |                |               |               |                 |     |               |               |                  |                  |                  |                  |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|-----------------------------------------------|----------------------------------------------|-----------------------------------------------|-----------------|----|-------------------|-------------------|--------|----------------|---------------|---------------|-----------------|-----|---------------|---------------|------------------|------------------|------------------|------------------|
| Physical Hazards             | Physical hazards include all hazards that are not atmospheric hazards, including: explosives (other than explosive atmospheres); mechanical, electrical, hydraulic and pneumatic energy; radiation; temperature extremes; engulfment; noise; inwardly converging surfaces; and chemicals that can cause death or serious physical harm through skin or eye contact (rather than through inhalation).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                              |                                               |                                              |                                               |                 |    |                   |                   |        |                |               |               |                 |     |               |               |                  |                  |                  |                  |
| Safe Condition (Atmospheric) | <p>Safe Condition within a confined space is defined as having an atmosphere within the following ranges:</p> <table><tr><th>Atmosphere Monitored</th><th>Abbreviation</th><th>Hazardous Atmosphere Acceptable Range/Levels</th><th>Hazardous Environment Acceptable Range/Levels</th></tr><tr><td>Carbon Monoxide</td><td>CO</td><td>*Less than 25 ppm</td><td>*Less than 25 ppm</td></tr><tr><td>Oxygen</td><td>O<sub>2</sub></td><td>19.5% - 23.5%</td><td>19.5% - 22.0%</td></tr><tr><td>Combustible Gas</td><td>LEL</td><td>Less than 10%</td><td>Less than 10%</td></tr><tr><td>Hydrogen Sulfide</td><td>H<sub>2</sub>S</td><td>*Less than 1 ppm</td><td>*Less than 1 ppm</td></tr></table> <p><i>*Atmospheres that are not within the acceptable limits are shown above. The air monitoring instrument shall be set to alarm at the Permissible Exposure Limit (PEL) or the Threshold Limit Value (TLV) for toxic gases.</i></p> | Atmosphere Monitored                         | Abbreviation                                  | Hazardous Atmosphere Acceptable Range/Levels | Hazardous Environment Acceptable Range/Levels | Carbon Monoxide | CO | *Less than 25 ppm | *Less than 25 ppm | Oxygen | O <sub>2</sub> | 19.5% - 23.5% | 19.5% - 22.0% | Combustible Gas | LEL | Less than 10% | Less than 10% | Hydrogen Sulfide | H <sub>2</sub> S | *Less than 1 ppm | *Less than 1 ppm |
| Atmosphere Monitored         | Abbreviation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Hazardous Atmosphere Acceptable Range/Levels | Hazardous Environment Acceptable Range/Levels |                                              |                                               |                 |    |                   |                   |        |                |               |               |                 |     |               |               |                  |                  |                  |                  |
| Carbon Monoxide              | CO                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | *Less than 25 ppm                            | *Less than 25 ppm                             |                                              |                                               |                 |    |                   |                   |        |                |               |               |                 |     |               |               |                  |                  |                  |                  |
| Oxygen                       | O <sub>2</sub>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 19.5% - 23.5%                                | 19.5% - 22.0%                                 |                                              |                                               |                 |    |                   |                   |        |                |               |               |                 |     |               |               |                  |                  |                  |                  |
| Combustible Gas              | LEL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Less than 10%                                | Less than 10%                                 |                                              |                                               |                 |    |                   |                   |        |                |               |               |                 |     |               |               |                  |                  |                  |                  |
| Hydrogen Sulfide             | H <sub>2</sub> S                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | *Less than 1 ppm                             | *Less than 1 ppm                              |                                              |                                               |                 |    |                   |                   |        |                |               |               |                 |     |               |               |                  |                  |                  |                  |

## 1.5 Authorizing Policies

This program is authorized by BPA Policy 420-1 Occupational Safety and Health.

## 1.6 Roles and Responsibilities

### Managers & Supervisors:

- Ensure all spaces meeting the definition of a Permit-Required Confined Space within area of responsibility are identified and labeled in accordance with this program.
- Ensure all employees are informed as to the locations and hazards associated with all Permit-Required Confined Spaces in their respective work areas.

|                           |                                   |                                                                |                      |                             |  |
|---------------------------|-----------------------------------|----------------------------------------------------------------|----------------------|-----------------------------|--|
| Organization<br><b>NF</b> |                                   | Title/Subject<br><b>Permit Required Confined Space Program</b> |                      | Unique ID<br><b>420-601</b> |  |
| Owner<br><b>CSO</b>       | Approved by<br><b>John Cannon</b> | Date<br><b>9/14/2020</b>                                       | Version<br><b>V1</b> | Page<br><b>6</b>            |  |

- Ensure all work requiring access to a PRCS is conducted under the supervision of a trained and qualified Entry Supervisor.
- Enable resourcing and acquisition of equipment and PPE needed to perform confined space entry.
- Ensure resourcing of Confined Space training that satisfies 29 CFR 1910.146 and 1926 Subpart AA. Utilize the Safety Organization to evaluate quality and scope of training programs.

#### **Competent Persons:**

- Conduct confined space hazard analysis as described in this program.
- Maintain knowledge and understanding of confined space regulations, policies and procedures applicable to BPA.
- Through hazard analysis, make determinations as to the appropriate confined space entry procedure needed.
- Serve as, or designate another qualified employee, the Entry Supervisor for PRCS entry.

#### **Entry Supervisors:**

- Complete Permit-Required Confined Space, Entry Supervisor level training (satisfying 29 CFR 1910.146 and 1926 Subpart AA) no less than every 3 years.
- Maintain understanding of the current confined space standards, hazards associated with confined spaces, proper use of Personal Protective Equipment (PPE), and rescue equipment.
- Understand the roles and responsibilities of those involved in a PRCS entry (entry supervisor, attendant, entrant, and rescue team).
- Ensure hazards are identified and mitigated appropriately prior to any PRCS entry.
- Ensure all employees assigned confined space entry roles are current with training and knowledge required to execute their responsibilities.

#### **Attendants:**

- Complete Permit-Required Confined Space, Entry Supervisor level training (satisfying 29 CFR 1910.146 and 1926 Subpart AA) no less than every 3 years. This training must include responsibilities of Attendants.
- Remains present during all entries into any area classified as a PRCS
- Must not perform any duties/activities that interfere with his/her ability to monitor and protect authorized entrants

|                           |                                   |                                                                |  |                             |                  |
|---------------------------|-----------------------------------|----------------------------------------------------------------|--|-----------------------------|------------------|
| Organization<br><b>NF</b> |                                   | Title/Subject<br><b>Permit Required Confined Space Program</b> |  | Unique ID<br><b>420-601</b> |                  |
| Owner<br><b>CSO</b>       | Approved by<br><b>John Cannon</b> | Date<br><b>9/14/2020</b>                                       |  | Version<br><b>V1</b>        | Page<br><b>7</b> |

- Responsible for knowing the hazards that may be encountered during entry, including information on the mode, signs or symptoms, and consequences of the exposure.
- Maintains awareness of possible behavioral effects of hazardous atmospheres on entrants.
- Continuously maintains an accurate count of entrants in the PRCS and updates the permit as authorized personnel enter and leave the PRCS.
- Remains outside the PRCS during entry operations.
- Communicates with entrants to monitor their status and alerts entrants of the need to evacuate as necessary.
- Monitors activities inside and outside the PRCS and orders entrants to evacuate if prohibited conditions develop.

**Authorized Entrants:**

- Complete Permit-Required Confined Space, Entry Supervisor level training (satisfying 29 CFR 1910.146 and 1926 Subpart AA) no less than every 3 years. This training must include responsibilities of Authorized Entrants.
- Responsible for knowing the hazards that may be encountered during entry, including information on the mode, signs or symptoms, and consequences of the exposure.
- Properly uses equipment needed to safely enter confined spaces and perform the necessary task.
- Communicates with the attendant, as necessary, to enable the attendant to monitor entrant status and alert entrants of the need to evacuate the space should a prohibited condition develop.

**Non-Entry Rescue Team:**

- Complete annual PRCS rescue training to include hands-on simulation exercise of an actual rescue.
- Maintain understanding of the current confined space standards, hazards associated with confined spaces, proper use of Personal Protective Equipment (PPE), and rescue equipment.
- Maintain current certifications in basic first aid, CPR, Automated External Defibrillator (AED) and Blood borne Pathogens.
- Maintain enrollment in the Respiratory Protection Program with a clearance to wear any needed respirator for rescue.

**Safety Organization:**

|                    |                            |                                                         |                   |                      |           |
|--------------------|----------------------------|---------------------------------------------------------|-------------------|----------------------|-----------|
| Organization<br>NF |                            | Title/Subject<br>Permit Required Confined Space Program |                   | Unique ID<br>420-601 |           |
| Owner<br>CSO       | Approved by<br>John Cannon |                                                         | Date<br>9/14/2020 | Version<br>V1        | Page<br>8 |

- Maintains this program in accordance with Safety Management Systems standards of continuous improvement.
- Receives and reviews completed *Confined Space Assessments* and *Permit Required Confined Space Entry Forms*.
- Provide technical advice regarding confined space hazards and mitigation strategies to employees with roles in confined space entry.
- Conduct job observations and quality assurance reviews of documentation to establish the effectiveness of this program and to recommend improvements.
- Evaluate Confined Space training programs for compliance with 29 CFR 1910.146 and 1926 Subpart AA.

#### **Employees:**

- Maintain knowledge of the locations and associated hazards of confined spaces within their work area.
- Do not access any confined space outside of the conditions set in this program.
- Report all identified hazards associated with confined spaces through the Near-Hit/Safety Concern Program.

## **2. Identification of Permit-Required Confined Spaces**

### **2.1 Policy**

BPA personnel shall consider all confined spaces to be Permit-Required Confined Spaces until reclassified as described in Section 2.3 of this program. Only qualified employees as described in this program may enter a PRCS.

### **2.2 Identifying Permit-Required Confined Spaces**

Managers and supervisors will ensure that all PRCS within their area of responsibility are identified and marked with appropriate signage. Employees must also be informed of the location of these spaces and the possible dangers associated with them. PRCS's will be labeled with signage featuring "DANGER" in white letters on a red background with language referring to the permit requirement as shown below:



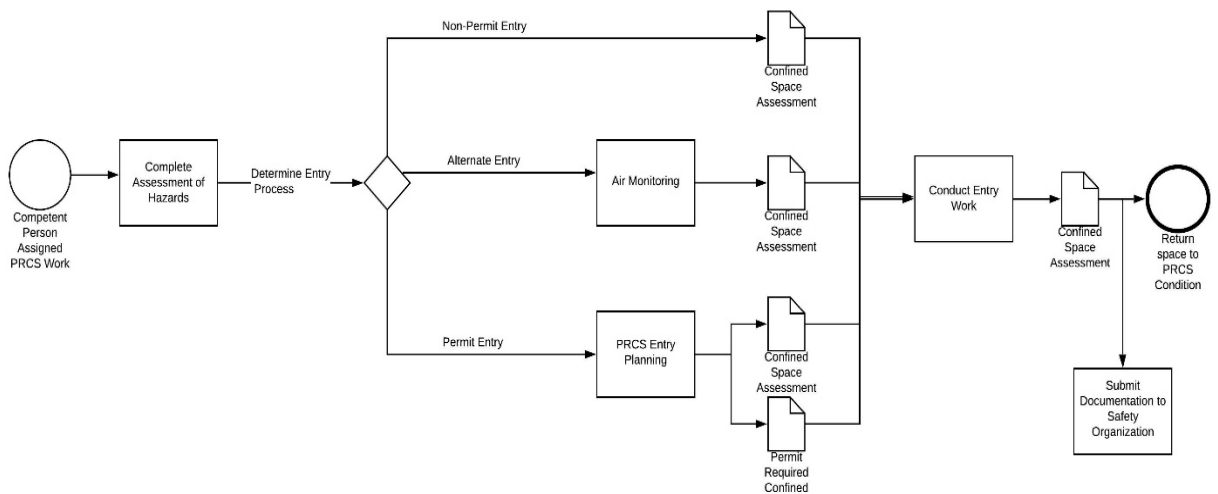
|                           |                                   |                                                                |  |                             |                  |
|---------------------------|-----------------------------------|----------------------------------------------------------------|--|-----------------------------|------------------|
| Organization<br><b>NF</b> |                                   | Title/Subject<br><b>Permit Required Confined Space Program</b> |  | Unique ID<br><b>420-601</b> |                  |
| Owner<br><b>CSO</b>       | Approved by<br><b>John Cannon</b> | Date<br><b>9/14/2020</b>                                       |  | Version<br><b>V1</b>        | Page<br><b>9</b> |

## 2.3 Controlling Access to Permit-Required Confined Spaces

All employees who may encounter a PRCS through the course of their work duties must be informed as to the location and dangers presented by the space. Managers and supervisors must take appropriate steps to restrict inadvertent access to such spaces and to deter unauthorized entry.

## 3. Permit-Required Confined Space Entry Planning

### 3.1 Process Map



### 3.2 Designation of a Competent Person

When entry into a space designated as PRCS is required as part of operations or maintenance, the supervisor or manager with authority over the space must designate a Competent Person to evaluate the space. It is recommended that a qualified Entry Supervisor who will lead the entry is designated to assess the space.

### 3.3 Conducting the PRCS Hazard Analysis

3.3.1 The Competent Person (CP) must conduct a PRCS Hazard Analysis utilizing BPA F 5480.10e– Confined Space Assessment. The CP must include on the form the following information as part of the analysis:

- Site Location or description
- Date of Assessment
- Purpose of Entry
- Type of work to be performed

|                           |                                   |                                                                |  |                             |                   |
|---------------------------|-----------------------------------|----------------------------------------------------------------|--|-----------------------------|-------------------|
| Organization<br><b>NF</b> |                                   | Title/Subject<br><b>Permit Required Confined Space Program</b> |  | Unique ID<br><b>420-601</b> |                   |
| Owner<br><b>CSO</b>       | Approved by<br><b>John Cannon</b> | Date<br><b>9/14/2020</b>                                       |  | Version<br><b>V1</b>        | Page<br><b>10</b> |

- The name of the CP and a contact phone number

3.3.2 The CP must indicate as part of their assessment any hazards that may present during work in the space. The CP may utilize technical specifications, previous work reports, and subject matter experts to identify all hazards for a particular space.

3.3.3 The CP is not permitted to physically enter the space as part of the hazard analysis. If entry is required to conduct a thorough hazard analysis the entry shall be done as a separate PRCS Entry.

### 3.4 Determining Hazard Controls

Upon completion of the hazard analysis, the CP must identify hazard controls that mitigate or eliminate the indicated hazards. Controls listed on BPA F 5480.10e – *Confined Space Assessment* are numbered to correspond to the hazard identified in the previous section. All hazards identified must have controls selected as part of the PRCS process.

### 3.5 Determining Space Designation

3.5.1 Upon completion of identifying corresponding hazard controls for all potential hazards, the CP must make a determination on the entry requirements of the space. The Confined Space Assessment form guides the CP towards the most appropriate of three entry methods:

- Non-Permit Entry (Section 4)
- Alternate Entry (Section 5)
- Permit Entry (Section 6)

3.5.2 The CP may indicate a Non-Permit Entry on the Confined Space Assessment when all of the following criteria are met:

- No Permit Required Hazards (shaded in gray on the *Confined Space Assessment*) were identified.
- No known or potential atmospheric hazards (shaded in yellow) are present in the space.
- There are no physical hazards **OR** all physical hazards identified are able to be eliminated or isolated without entering the space.

|                           |                                   |                                                                |  |                             |                   |
|---------------------------|-----------------------------------|----------------------------------------------------------------|--|-----------------------------|-------------------|
| Organization<br><b>NF</b> |                                   | Title/Subject<br><b>Permit Required Confined Space Program</b> |  | Unique ID<br><b>420-601</b> |                   |
| Owner<br><b>CSO</b>       | Approved by<br><b>John Cannon</b> | Date<br><b>9/14/2020</b>                                       |  | Version<br><b>V1</b>        | Page<br><b>11</b> |

3.5.3 Spaces, in which the only hazard is a potential or actual hazardous atmosphere and the CP can demonstrate hazard control by continuous air monitoring and ventilation (shaded in yellow on the Confined Space Assessment), may be accessed as Alternate Entry.

3.5.4 Any space with a potential hazard or control method notated on the Confined Space Assessment that is shaded in gray, requires a Permit Entry. Additionally, any entry into the following spaces requires a Permit Entry:

- Reactors
- Transformers
- GIS Equipment

3.5.5 The CP must indicate the most appropriate entry method on the Confined Space Assessment form.

|                           |                                   |                                                                |  |                             |                   |
|---------------------------|-----------------------------------|----------------------------------------------------------------|--|-----------------------------|-------------------|
| Organization<br><b>NF</b> |                                   | Title/Subject<br><b>Permit Required Confined Space Program</b> |  | Unique ID<br><b>420-601</b> |                   |
| Owner<br><b>CSO</b>       | Approved by<br><b>John Cannon</b> | Date<br><b>9/14/2020</b>                                       |  | Version<br><b>V1</b>        | Page<br><b>12</b> |

## 4. Non-Permit Confined Space Entry

### 4.1 Non-Permit Entry Planning

- 4.1.1 When the CP has indicated a non-permit entry on The Confined Space Assessment form, they must sign and date to confirm that determination. This action reclassifies the space as Non-Permit for the duration of the work to be performed.
- 4.1.2 The signed form must be forwarded to the District Safety Manager. Providing the form to the Safety Manager prior to work starting is an opportunity for additional review but it is not required if the operation timetable is prohibitive.
- 4.1.3 The CP must conduct a Job Hazard Analysis to address any exposures or hazards that may be present during work. The Accident Prevention Manual (APM) J-1 guides workers through the Risk Management Process for work tasks.

### 4.2 Non-Permit Entry

- 4.2.1 The CP must ensure that all workers involved in the operation are informed of the hazards related to the non-permit entry and must post the signed Confined Space Assessment at a conspicuous location at the worksite.
- 4.2.2 The CP must monitor for hazards arising during work that require procedures consistent with a PRCS. When such hazards are encountered, each employee must exit immediately and the CP must reevaluate the space.

### 4.3 Non-Permit Entry Termination

- 4.3.1 At the conclusion of the work performed, the CP must ensure any previous hazard controls, restrictions to entry and labeling/signage is returned to the pre-work state.
- 4.3.2 The space is considered a Permit-Required Confined Space at the conclusion of the non-permit entry.

## 5. Alternate Entry Process

### 5.1 Alternate Entry Planning

|                           |                                   |                                                                |  |                             |                   |
|---------------------------|-----------------------------------|----------------------------------------------------------------|--|-----------------------------|-------------------|
| Organization<br><b>NF</b> |                                   | Title/Subject<br><b>Permit Required Confined Space Program</b> |  | Unique ID<br><b>420-601</b> |                   |
| Owner<br><b>CSO</b>       | Approved by<br><b>John Cannon</b> | Date<br><b>9/14/2020</b>                                       |  | Version<br><b>V1</b>        | Page<br><b>13</b> |

5.1.1 When the CP has indicated an Alternate Entry on *The Confined Space Assessment* form, they must sign and date to confirm that determination. This action reclassifies the space as Alternate Entry for the duration of the work to be performed.

5.1.2 The signed form must be forwarded to the District Safety Manager. Providing the form to the Safety Manager prior to work starting is an opportunity for additional review but it is not required if the operation timetable is prohibitive.

5.1.3 The CP must ensure the following objectives prior to entry as part of planning an Alternate Entry:

- Eliminate unsafe conditions before removing entrance covers.
- Promptly guard openings with railing or other temporary barriers.
- Test the internal atmosphere with a calibrated, four gas direct reading air monitor with hydrogen sulfide detection capability.
- Verify all ventilation systems are working properly.

5.1.4 The CP must designate themselves or another qualified individual to test, calibrate and operate any air monitoring equipment.

## 5.2 Alternate Entry

5.2.1 The CP must ensure that all workers involved in the operation are informed of the hazards related to the alternate permit entry and must post the signed *Confined Space Assessment* at a conspicuous location at the worksite.

5.2.2 The designated employee ensures that air monitoring is conducted no less than every two (2) hours for the duration of work.

5.2.3 The designated employee must ensure all atmospheric air monitoring is logged onto the *Confined Space Assessment* form.

5.2.4 The CP must ensure that all entrants into the space are afforded the opportunity to review atmospheric testing results.

5.2.5 Any employee who observes any of the following conditions must inform the CP and initiate an evacuation of all personnel in the space:

- Detection of a hazardous atmosphere by air-monitoring instrumentation.
- Failure of a direct-reading instrument.
- Any failure of the ventilation system.

|                           |                                   |                                                                |  |                             |                   |
|---------------------------|-----------------------------------|----------------------------------------------------------------|--|-----------------------------|-------------------|
| Organization<br><b>NF</b> |                                   | Title/Subject<br><b>Permit Required Confined Space Program</b> |  | Unique ID<br><b>420-601</b> |                   |
| Owner<br><b>CSO</b>       | Approved by<br><b>John Cannon</b> | Date<br><b>9/14/2020</b>                                       |  | Version<br><b>V1</b>        | Page<br><b>14</b> |

- Introduction of a hazard, a hazard develops, or conditions unexpectedly change within the space.

### 5.3 Alternate Entry Termination

- 5.3.1 At the conclusion of the work performed, the CP must ensure any previous hazard controls, restrictions to entry and labeling/signage is returned to the pre-work state.
- 5.3.2 The space is considered a Permit-Required Confined Space at the conclusion of the non-permit entry.
- 5.3.3 The CP must ensure all testing data is accurately recorded on the Confined Space Assessment form and forward a copy to the District Safety Manager.

## 6. Permit-Required Confined Space Entry

### 6.1 Permit-Required Confined Space Entry Planning

- 6.1.1 The CP who conducted the initial hazard assessment must designate a qualified Entry Supervisor to lead the PRCS Entry. The CP may designate themselves to act in that role if they are qualified to fulfil the responsibilities.
- 6.1.2 The Entry Supervisor must complete the *Permit Required Confined Space Form BPA F 5480.10a*. The Entry Supervisor must indicate:
- The start time and date, duration, and stop time and date of the permit.
  - Specify the method and procedures for communication during work.
  - Specify which of the approved rescue plans on the form will be utilized and coordinate with those rescue resources.
  - List the names of all rescue team members assigned to the operation.
  - List any special hazards, information, procedures or conditions relevant to rescue operations.

|                           |                                   |                                                                |  |                             |                   |
|---------------------------|-----------------------------------|----------------------------------------------------------------|--|-----------------------------|-------------------|
| Organization<br><b>NF</b> |                                   | Title/Subject<br><b>Permit Required Confined Space Program</b> |  | Unique ID<br><b>420-601</b> |                   |
| Owner<br><b>CSO</b>       | Approved by<br><b>John Cannon</b> | Date<br><b>9/14/2020</b>                                       |  | Version<br><b>V1</b>        | Page<br><b>15</b> |

6.1.3 The Entry Supervisor must complete all pre-entry procedures listed on the *Permit Required Confined Space Form* to indicate implemented controls and those that are not applicable to the entry.

6.1.4 The entry supervisor must designate an individual to calibrate, operate and record results from four gas direct reading air monitor with hydrogen sulfide detection capability.

6.1.5 The entry supervisor must designate a team consisting of at least one PRCs Attendant and an appropriate number of PRCs Authorized Entrants. The names of all team members must be recorded on the *Permit Required Confined Space Form*.

6.1.6 The Entry Supervisor must establish a retrieval system for personnel entering the PRCs. The retrieval system, must satisfy the following requirements:

- Allow for each entrant to use a chest or full-body harness with a retrieval line attached. Due to the narrow dimensions of some manways, wristlets are required to complete a vertical rescue of workers from these spaces.
- The end of each retrieval line must be attached to a mechanical device or a fixed point outside the space. The approved mechanical devices or fixed points are a tripod, davit arm, or crane to effect the rescue.
- A mechanical device for aiding exit will be made available for spaces more than five feet deep.
- The retrieval system will be applicable to the space (horizontal/vertical).
- The retrieval system does not, through its design, increase the overall risk of the entry.

6.1.7 The Entry Supervisor must ensure identified equipment needed for safe entry is available, in satisfactory condition, and implemented appropriately. This equipment may include:

- Testing and monitoring equipment
- Ventilating Equipment
- Communication Equipment
- PPE
- Barriers and shields
- Ladders for ingress and egress
- Rescue Equipment

|                           |                                   |                                                                |                          |                             |                   |
|---------------------------|-----------------------------------|----------------------------------------------------------------|--------------------------|-----------------------------|-------------------|
| Organization<br><b>NF</b> |                                   | Title/Subject<br><b>Permit Required Confined Space Program</b> |                          | Unique ID<br><b>420-601</b> |                   |
| Owner<br><b>CSO</b>       | Approved by<br><b>John Cannon</b> |                                                                | Date<br><b>9/14/2020</b> | Version<br><b>V1</b>        | Page<br><b>16</b> |

- Other Equipment as needed

6.1.8 The Entry Supervisor will sign the Permit Required Confined Space Form, thereby approving entry.

6.1.9 The signed form must be forwarded to the District Safety Manager. Providing the form to the Safety Manager prior to work starting is an opportunity for additional review but it is not required if the operation timetable is prohibitive.

6.1.10 The Entry Supervisor must post the completed and approved permit in a conspicuous location near the PRCS entrance for the duration of work.

## 6.2 Permit Required Confined Space Entry

6.2.1 Upon posting of the completed and signed *Permit Required Confined Space Form*, the Entry Supervisor may initiate entry into the PRCS. During entry the Entry Supervisor must ensure the following:

- Direct reading air monitoring equipment is used to detect hazardous atmospheres prior to and during the PRCS entry.
- Only four gas direct reading air monitors with hydrogen sulfide detection capability are utilized.
- All designated PRCS team members are afforded the opportunity to observe and review test results.
- All personnel involved in the operation are trained and qualified for the tasks and equipment under their responsibility.
- Hazard controls remain in place and are effective.

6.2.2 The Entry Supervisor must verify that conditions in the PRCS are acceptable for entry throughout the duration of an authorized entry. The PRCS must be continuously monitored with a calibrated, direct reading air monitoring device for atmospheric hazards and the results recorded every two hours on the form.

6.2.3 Any employee who observes any of the following conditions must inform the Entry Supervisor and initiate an evacuation of all personnel in the space:

- Detection of a hazardous atmosphere by air-monitoring instrumentation.
- Failure of a direct-reading instrument.
- Any failure of the ventilation system.

|                           |                                   |                                                                |  |                             |                   |
|---------------------------|-----------------------------------|----------------------------------------------------------------|--|-----------------------------|-------------------|
| Organization<br><b>NF</b> |                                   | Title/Subject<br><b>Permit Required Confined Space Program</b> |  | Unique ID<br><b>420-601</b> |                   |
| Owner<br><b>CSO</b>       | Approved by<br><b>John Cannon</b> | Date<br><b>9/14/2020</b>                                       |  | Version<br><b>V1</b>        | Page<br><b>17</b> |

- Introduction of a hazard, a hazard develops, or conditions unexpectedly change within the space.

6.2.4 The permit may be temporarily suspended by the Entry Supervisor to accommodate the logistics of the work. Prior to any re-entry of the space the Entry Supervisor must ensure the following:

- The atmosphere shall be re-tested and results logged on the permit.
- The entry supervisor shall verify that all precautions and other measures called for on the permit are still in effect.
- Only work originally approved on the permit shall be conducted in the confined space.

### 6.3 PRCS Termination

6.3.1 At the conclusion of the work performed, the Entry Supervisor must ensure any previous hazard controls, restrictions to entry and labeling/signage is returned to the pre-work state.

6.3.2 At the conclusion of all permitted work, the Entry Supervisor must ensure the *Permit Required Confined Space Form* is forwarded to the District Safety Manager.

## 7. References

### 7.1 Regulations

OSHA – 29 CFR 1910.134, Respiratory Protection Program

OSHA - 29 CFR 1910.146, Permit-Required Confined Spaces

OSHA – 29 CFR 1910.1000, Air Contaminants

OSHA - 29 CFR 1926.146, Subpart AA, Confined Spaces in Construction

### 7.2 Standards

Accident Prevention Manual (APM)), Rule C-3, Confined Spaces

BPA Safety & Occupational Health Manual, 420-660-01, Fall Protection Handbook

BPA-WS-11-7 Rescue Policy, BPA-WS-11-7 – Personnel Safety Procedures Rescue Policy

ANSI Z117, Safety Requirements for Confined Spaces Guidelines

|                           |                                   |                                                                |  |                             |                   |
|---------------------------|-----------------------------------|----------------------------------------------------------------|--|-----------------------------|-------------------|
| Organization<br><b>NF</b> |                                   | Title/Subject<br><b>Permit Required Confined Space Program</b> |  | Unique ID<br><b>420-601</b> |                   |
| Owner<br><b>CSO</b>       | Approved by<br><b>John Cannon</b> | Date<br><b>9/14/2020</b>                                       |  | Version<br><b>V1</b>        | Page<br><b>18</b> |

### 7.3 Training

Confined Space training must meet the requirements listed in 29 CFR 1910.146 and 1926 Subpart AA.

All employees with Confined Space roles and responsibilities will be trained to the Entry Supervisor level.

## 8. Review

### 8.1 Review Timeline

This program document will be reviewed and revised in accordance with the SMS. All Confined Space Assessments and Permit Required Confined Space Forms will be reviewed within one year submission.

## 9. Document Change Control

This chart contains a history of the revisions and reviews made to this document.

| Version Number | Date of Issue | Author/Reviewer | Brief Description of Change/Review |
|----------------|---------------|-----------------|------------------------------------|
|                |               |                 |                                    |

|                           |                                   |                                                                |                          |                             |                   |
|---------------------------|-----------------------------------|----------------------------------------------------------------|--------------------------|-----------------------------|-------------------|
| Organization<br><b>NF</b> |                                   | Title/Subject<br><b>Permit Required Confined Space Program</b> |                          | Unique ID<br><b>420-601</b> |                   |
| Owner<br><b>CSO</b>       | Approved by<br><b>John Cannon</b> |                                                                | Date<br><b>9/14/2020</b> | Version<br><b>V1</b>        | Page<br><b>19</b> |

# 10. Confined Space Assessment/Permit Required Confined Space Form

BPA F 5480.10e

(09-2020)

(Previous editions unusable)

U.S. DEPARTMENT OF ENERGY-BONNEVILLE POWER ADMINISTRATION(BPA)

## CONFINED SPACE ASSESSMENT

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                              |                                                                                                                                                                                                                                                                                                                                                                                                                         |                                              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| Site location or description:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                              | Date:                                                                                                                                                                                                                                                                                                                                                                                                                   |                                              |
| Purpose of entry:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                              | Type of Work:                                                                                                                                                                                                                                                                                                                                                                                                           |                                              |
| Competent Person or CS Supervisor Evaluating:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                              | Work Phone Number:                                                                                                                                                                                                                                                                                                                                                                                                      |                                              |
| <p><i>A Competent Person must evaluate the space using the Hazard Analysis Below</i></p> <ul style="list-style-type: none"> <li>If all hazards cannot be eliminated by checking the corresponding method below or hazards in the grey box are checked, the space is a Permit Required Confined Space BPA F 5480.10e (09-2020) must be filled out.</li> <li>If the only hazard posed by the permit space is an actual or potential hazardous atmosphere and all other hazards have been eliminated, the space can be classified as an <b>Alternate Procedures</b> space. Air monitoring and ventilation are required for these spaces.</li> <li>If there are no hazards or all hazards have been eliminated by checking the corresponding box below, then the space can be reclassified as a Non-Permit Confined Space.</li> </ul> |                                              |                                                                                                                                                                                                                                                                                                                                                                                                                         |                                              |
| <b>HAZARD ANALYSIS</b><br><input type="checkbox"/> (1) Potential/Known Hazardous Atmosphere<br><input type="checkbox"/> (2) Potential Explosive Hazard<br><input type="checkbox"/> (3) Electrical Hazards<br><input type="checkbox"/> (4) Excessive Heat/Cold<br><input type="checkbox"/> (5) High Noise (> 85 dBA)<br><input type="checkbox"/> (6) Fall hazard from height within<br><input type="checkbox"/> (7) Slippery or uneven surface                                                                                                                                                                                                                                                                                                                                                                                     |                                              | <input type="checkbox"/> (8) Chemical exposure<br><input type="checkbox"/> (9) Birds / Animals / Insects / Plants<br><input type="checkbox"/> (10) Heavy equipment operating<br><input type="checkbox"/> (11) Vehicle traffic near opening<br><input type="checkbox"/> (12) Water or equipment hazard<br><input type="checkbox"/> (13) Moving mechanical equipment<br><input type="checkbox"/> (14) Inadequate lighting |                                              |
| <b>PERMIT REQUIRED HAZARDS</b><br><input type="checkbox"/> (15) Atmosphere Requiring Respirator<br><input type="checkbox"/> (16) Known Explosive Atmosphere<br><input type="checkbox"/> (17) Welding/Fire Hazard<br><input type="checkbox"/> (18) Raw Sewage/Decaying Material<br><input type="checkbox"/> (19) Entanglement Hazard or Difficult Rescue<br><input type="checkbox"/> (20) Electrical Work without LOTO<br><input type="checkbox"/> (21) Work on Energized/Pressurized Systems                                                                                                                                                                                                                                                                                                                                      |                                              |                                                                                                                                                                                                                                                                                                                                                                                                                         |                                              |
| <b>CORRESPONDING METHODS TO CONTROL OR ELIMINATE OR HAZARDS ABOVE</b><br><input type="checkbox"/> (1) Air Monitoring/Ventilation (Control Only)<br><input type="checkbox"/> (2) Air Monitoring/Ventilation<br><input type="checkbox"/> (3) Lockout/Tagout of Electrical<br><input type="checkbox"/> (4) Heaters/Administrative Controls/Clothing<br><input type="checkbox"/> (5) Hearing protection<br><input type="checkbox"/> (6) Fall Protection<br><input type="checkbox"/> (7) Cleanup or Engineering Control                                                                                                                                                                                                                                                                                                                |                                              |                                                                                                                                                                                                                                                                                                                                                                                                                         |                                              |
| <input type="checkbox"/> (8) Hazard Elimination per SDS<br><input type="checkbox"/> (9) Removal of natural hazard<br><input type="checkbox"/> (10) Equipment Control<br><input type="checkbox"/> (11) Traffic Control<br><input type="checkbox"/> (12) Water removal/LOTO<br><input type="checkbox"/> (13) Lockout/Tagout<br><input type="checkbox"/> (14) Lighting                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                              |                                                                                                                                                                                                                                                                                                                                                                                                                         |                                              |
| <input type="checkbox"/> (15) Monitoring/Ventilation <input type="checkbox"/> Respirators<br><input type="checkbox"/> (16) Monitoring/Ventilation - Intrinsically Safe<br><input type="checkbox"/> (17) Fire Extinguishers/ (Hot Work Permit)<br><input type="checkbox"/> (18) PPE for Sewage<br><input type="checkbox"/> (19) Rescue plan/Retrieval equipment<br><input type="checkbox"/> (20) Qualified person/Arc Flash protection<br><input type="checkbox"/> (21) Qualified person and work policy                                                                                                                                                                                                                                                                                                                           |                                              |                                                                                                                                                                                                                                                                                                                                                                                                                         |                                              |
| <b>SPACE DESIGNATED AS:</b> <input type="checkbox"/> NON-PERMIT SPACE <input type="checkbox"/> <b>ALTERNATE PROCEDURES ENTRY</b><br><input type="checkbox"/> <b>PERMIT REQUIRED CONFINED SPACE – FILL OUT REVERSE SIDE</b><br><small>(IF ANY HAZARDS CANNOT BE ELIMINATED OR HAZARDS IN GRAY BOX ARE CHECKED, THE SPACE IS AUTOMATICALLY PERMIT REQUIRED AND PAGE 2 MUST BE COMPLETED)</small>                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                              |                                                                                                                                                                                                                                                                                                                                                                                                                         |                                              |
| Date: _____ Signature of Supervisor for Non- Permit or Alternate Procedures Entry: _____<br><input type="checkbox"/> Air Monitoring Required (record below) <input type="checkbox"/> Continuous Ventilation (Optional calculate below) <input type="checkbox"/> Natural Ventilation (Open Spaces only)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                              |                                                                                                                                                                                                                                                                                                                                                                                                                         |                                              |
| <p><i>Note: Employees should notify an outside contact when entering an ALTERNATE PROCEDURES space and check in at regularly scheduled intervals with the outside contact. (Such as every _____ minutes and when exiting the space).</i></p> <p><b>Exit space immediately if there is a failure of a direct reading instrument, failure of ventilation system, detection of a hazardous atmosphere, introduction of a new hazard, a hazard develops, entrant shows signs of exposure, or conditions change in the space.</b></p>                                                                                                                                                                                                                                                                                                  |                                              |                                                                                                                                                                                                                                                                                                                                                                                                                         |                                              |
| <input type="checkbox"/> <b>AIR MONITORING FOR ALTERNATE PROCEDURES</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                              | <input type="checkbox"/> <b>AIR MONITORING FOR PERMIT REQUIRED</b>                                                                                                                                                                                                                                                                                                                                                      |                                              |
| Gas Monitor Model:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Bump Test Today:                             | Days to Calibration:                                                                                                                                                                                                                                                                                                                                                                                                    |                                              |
| Substance:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Permissible Exposure Levels:                 | Continuous Air Monitoring Record Results ( _____ Minute Intervals):                                                                                                                                                                                                                                                                                                                                                     |                                              |
| All monitored levels must be below the PEL or within limits for safe entry conditions to be satisfied.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                              | Time:                                                                                                                                                                                                                                                                                                                                                                                                                   |                                              |
| Oxygen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Range: 19.5% to 23.5%                        |                                                                                                                                                                                                                                                                                                                                                                                                                         |                                              |
| LEL/LFL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Under 10%                                    |                                                                                                                                                                                                                                                                                                                                                                                                                         |                                              |
| H <sub>2</sub> S                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | PEL = 10 STEL = 15                           |                                                                                                                                                                                                                                                                                                                                                                                                                         |                                              |
| CO                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | PEL = 35 STEL = 200                          |                                                                                                                                                                                                                                                                                                                                                                                                                         |                                              |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                              |                                                                                                                                                                                                                                                                                                                                                                                                                         |                                              |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                              |                                                                                                                                                                                                                                                                                                                                                                                                                         |                                              |
| Optional Calculation:<br>Ventilation equipment and CFMs for recommended Air Exchanges Per Hour (AEPH)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Approximate Volume of Space in Feet (LxWxH): | Divide Volume of Space + BY 3 (For 20 AE/HR)                                                                                                                                                                                                                                                                                                                                                                            | = CFMs Needed for Air Exchanges Recommended: |

File Code: PE-53-17 Retention: 10 yrs

|                    |                            |                                                         |               |                      |  |
|--------------------|----------------------------|---------------------------------------------------------|---------------|----------------------|--|
| Organization<br>NF |                            | Title/Subject<br>Permit Required Confined Space Program |               | Unique ID<br>420-601 |  |
| Owner<br>CSO       | Approved by<br>John Cannon | Date<br>9/14/2020                                       | Version<br>V1 | Page<br>20           |  |

BPA F 5480.10a  
(09-2020)  
(Previous editions unusable)

## U.S. DEPARTMENT OF ENERGY-BONNEVILLE POWER ADMINISTRATION (BPA)

## PERMIT REQUIRED CONFINED SPACE FORM

| (THIS FORM MUST BE COMPLETED)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                 |                                                                                                                                                                                                                 |                                                                                                                                                                                                                       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Permit duration:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Permit Start Time/Date:                                                                                                                                                         | Permit Closed Time/Date:                                                                                                                                                                                        |                                                                                                                                                                                                                       |
| Communication procedures (including equipment):                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                 |                                                                                                                                                                                                                 |                                                                                                                                                                                                                       |
| RESCUE PLAN FOR PERMIT SPACES ONLY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                 |                                                                                                                                                                                                                 |                                                                                                                                                                                                                       |
| Rescue Team Number or Channel:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Rescue Team Contact Name:                                                                                                                                                       | Time and Date Rescue Team Notified:                                                                                                                                                                             |                                                                                                                                                                                                                       |
| - Non-entry rescue by attendant (entrant does not disconnect from system or have any significant entanglement hazards).<br>- Contracted previously qualified third party Rescue Team or Local Fire Department with Confined Space Technical Rescue team, available within reasonable response time, and who agrees to notify Supervisor if unavailable.<br>- Employer Rescue Team notified and available within reasonable response time. <input type="checkbox"/> Check here if team required at Entrance.<br>- Rescue Team ON SITE with supplied air respirators for known hazardous atmosphere entries. |                                                                                                                                                                                 |                                                                                                                                                                                                                 |                                                                                                                                                                                                                       |
| Rescue team names:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                 |                                                                                                                                                                                                                 |                                                                                                                                                                                                                       |
| Special hazards, information, procedures, or conditions for rescue:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                 |                                                                                                                                                                                                                 |                                                                                                                                                                                                                       |
| PRE-ENTRY PROCEDURES TO BE COMPLETED BEFORE ENTRY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                 |                                                                                                                                                                                                                 |                                                                                                                                                                                                                       |
| Area around space secured. (CS Signage posted and barriers placed around entrance.)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <input type="checkbox"/> Completed<br><input type="checkbox"/> NA                                                                                                               | Entrant in PPE/Respirator/Clothing marked below.                                                                                                                                                                | <input type="checkbox"/> Completed<br><input type="checkbox"/> NA                                                                                                                                                     |
| Energy Control Program in place. (LOTO) List Procedure or Reference:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <input type="checkbox"/> Completed<br><input type="checkbox"/> NA                                                                                                               | Emergency Escape Retrieval Equipment Setup. (Required at 5' Vertical Drop)<br><input type="checkbox"/> Tripod <input type="checkbox"/> Davit <input type="checkbox"/> Winch <input type="checkbox"/> Horizontal | <input type="checkbox"/> Completed<br><input type="checkbox"/> NA                                                                                                                                                     |
| Engulfment Line(s) Disconnected – LOTO – Water removed or controlled with pumps.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <input type="checkbox"/> Completed<br><input type="checkbox"/> NA                                                                                                               | Communications: <input type="checkbox"/> Visual <input type="checkbox"/> Verbal <input type="checkbox"/> Radio<br><input type="checkbox"/> Other: (Confirm adequate Lighting)                                   | <input type="checkbox"/> Completed<br><input type="checkbox"/> NA                                                                                                                                                     |
| Gas Lines - Purge, Flush or Vent Space - List Procedure or Reference:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <input type="checkbox"/> Completed<br><input type="checkbox"/> NA                                                                                                               | <b>HOT WORK PERMIT (REQUIRED FOR WELDING IN SPACE)</b><br>This document also serves as Hot Work Permit for hot work with flammables within 35' of work. Time Started: _____                                     |                                                                                                                                                                                                                       |
| Initial Air Monitoring recorded on page 1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <input type="checkbox"/> Completed<br><input type="checkbox"/> NA                                                                                                               | Fire Extinguisher Locations: _____                                                                                                                                                                              |                                                                                                                                                                                                                       |
| Ventilation of space before entry completed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <input type="checkbox"/> Completed<br><input type="checkbox"/> NA                                                                                                               | Name of Fire Watch: _____                                                                                                                                                                                       |                                                                                                                                                                                                                       |
| Chemical SDS reviewed and hazards addressed:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <input type="checkbox"/> Completed<br><input type="checkbox"/> NA                                                                                                               | Time of Fire Watch Completion: _____                                                                                                                                                                            |                                                                                                                                                                                                                       |
| Entrants must exit space immediately if there is a failure of a direct reading instrument, failure of ventilation system, detection of a hazardous atmosphere, introduction of a new hazard, a hazard develops, entrant shows signs of exposure, or conditions change.                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                 |                                                                                                                                                                                                                 |                                                                                                                                                                                                                       |
| Mark Entrant PPE required below:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                 |                                                                                                                                                                                                                 |                                                                                                                                                                                                                       |
| <input type="checkbox"/> Full body harness w/ dorsal "D" ring OR<br><input type="checkbox"/> Wristlets or similar retrieval attachment                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <input type="checkbox"/> Hardhat<br><input type="checkbox"/> Helmet<br><input type="checkbox"/> Boots                                                                           | <input type="checkbox"/> Hearing Protection<br><input type="checkbox"/> Air Purifying Respirator<br><input type="checkbox"/> SAR/SCBA                                                                           | <input type="checkbox"/> Goggles <input type="checkbox"/> Safety glasses<br><input type="checkbox"/> CPC <input type="checkbox"/> Faceshield<br><input type="checkbox"/> Gloves <input type="checkbox"/> Welding hood |
| Mark Attendant Required PPE Right:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <input type="checkbox"/> Full body harness <input type="checkbox"/> Hardhat <input type="checkbox"/> Gloves <input type="checkbox"/> Respirator <input type="checkbox"/> Other: |                                                                                                                                                                                                                 |                                                                                                                                                                                                                       |
| ATTENDANTS AND ENTRANTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                 |                                                                                                                                                                                                                 |                                                                                                                                                                                                                       |
| List lead Attendant:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | List Entrant(s):                                                                                                                                                                | Enter Time:                                                                                                                                                                                                     | Exit Time:                                                                                                                                                                                                            |
| List additional Attendants Below:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                 |                                                                                                                                                                                                                 |                                                                                                                                                                                                                       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                 |                                                                                                                                                                                                                 |                                                                                                                                                                                                                       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                 |                                                                                                                                                                                                                 |                                                                                                                                                                                                                       |
| SUPERVISOR                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                 |                                                                                                                                                                                                                 |                                                                                                                                                                                                                       |
| ENTRY SUPERVISOR SIGNATURE SIGNIFIES ALL CONDITIONS HAVE BEEN SATISFIED FOR ENTRY. Acceptable entry conditions for permit required spaces are all hazards controlled, continuous monitoring, ventilation systems functioning, and rescue planned.                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                 |                                                                                                                                                                                                                 |                                                                                                                                                                                                                       |
| Supervisor Name:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                 | Supervisor Phone:                                                                                                                                                                                               |                                                                                                                                                                                                                       |
| Supervisor Signature:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                 | Date and Time:                                                                                                                                                                                                  |                                                                                                                                                                                                                       |
| Return this form to Program Administrator. All confined space permits are to be kept on file for one year from date of use.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                 |                                                                                                                                                                                                                 |                                                                                                                                                                                                                       |

File Code: PE-53-17

Retention: 10 yrs

|                    |                                                         |                      |
|--------------------|---------------------------------------------------------|----------------------|
| Organization<br>NF | Title/Subject<br>Permit Required Confined Space Program | Unique ID<br>420-601 |
| Owner<br>CSO       | Approved by<br>John Cannon                              | Date<br>9/14/2020    |
|                    |                                                         | Version<br>V1        |
|                    |                                                         | Page<br>21           |